

(T11)

$$H_0: \xi \sim p_0(x) = 1 \{ (0, 1) \}$$

$$H_1: \xi \sim p_1(x) = \frac{e^{1-x}}{e-1} \{ (0, 1) \}$$

$$a) n=1$$

$$\rho = \frac{L_1}{L_0} = \frac{\frac{e^{1-x}}{e-1}}{1} \geq c \quad \text{T.O.} \quad G: x \leq A$$

$$P(x \leq A | H_0) = \alpha$$

$$\int_0^A 1 dx = A = \alpha \quad \Rightarrow \quad G: x \leq \alpha \quad \Rightarrow \quad \alpha_1 = \alpha$$

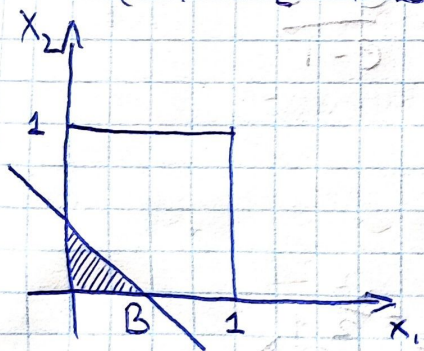
$$W = P(x \leq A | H_1) = \int_0^\alpha \frac{e}{e-1} e^{-x} dx = \frac{e}{e-1} (1 - e^{-\alpha})$$

$$\alpha_2 = 1 - W = 1 - \frac{e}{e-1} (1 - e^{-\alpha})$$

$$b) n=2$$

$$\rho = \frac{L_1}{L_0} = \frac{\frac{e^{1-x_1}}{e-1} \frac{e^{1-x_2}}{e-1}}{1 \cdot 1} = \frac{e^{2-x_1-x_2}}{(e-1)^2} \geq c \quad \Rightarrow \quad G: x_1 + x_2 \leq B$$

$$P(x_1 + x_2 \leq B | H_0) = \alpha$$



$$\iint_{\substack{x_1+x_2 \leq B \\ 0 < x_i \leq 1 \\ i=1,2}} 1 dx_1 dx_2 = \frac{B^2}{2} = \alpha \quad \Rightarrow \quad B = \sqrt{2\alpha}$$

$$G: x_1 + x_2 \leq \sqrt{2\alpha}$$

$$\alpha_1 = \alpha$$

$$W = P(x_1 + x_2 \leq B | H_1) = \iint_{\substack{x_1+x_2 \leq B \\ 0 < x_i \leq 1 \\ i=1,2}} \left(\frac{e}{e-1}\right)^2 e^{-x_1-x_2} dx_1 dx_2 =$$

$$= \left(\frac{e^2}{(e-1)^2}\right) \int_0^B e^{-x_1} dx_1 \int_0^{B-x_1} e^{-x_2} dx_2 = \left(\frac{e}{e-1}\right)^2 \int_0^B e^{-x_1} (1 - e^{-B+x_1}) dx_1 =$$

$$= \left(\frac{e}{e-1}\right)^2 (1 - e^{-B} - B e^{-B}) \quad \text{rde } B = \sqrt{2\alpha}$$

$$\alpha_2 = 1 - W$$

C) асимпт. крит.

$$P_2 \frac{L_1}{L_0} = \prod_{i=1}^n \frac{P_2(x_i)}{P_0(x_i)} \geq C$$

$$P_n P_2 = \sum_{i=1}^n P_n \frac{P_2(x_i)}{P_0(x_i)} \geq P_n C$$

$$\sum_{i=1}^n P_n P_2(x_i) - \sum_{i=1}^n P_n P_0(x_i) = \sum_{i=1}^n P_n \left(\frac{e^{1-x_i}}{e-1} \right) - 0 \geq P_n C$$

$$\sum_{i=1}^n (1-x_i) - n \ln(e-1) \geq P_n C$$

$$G: \sum_{i=1}^n x_i \leq A \Rightarrow \bar{x} \leq B$$

$$P(\bar{x} \leq B | H_0) = \alpha \quad ; \quad M_{\xi} = \frac{1}{2} \quad D_{\xi} = \frac{1}{12}$$

$$P(\bar{x} \leq B | H_0) = P\left(\frac{\bar{x} - \frac{1}{2}}{\sqrt{\frac{1}{12}}} \sqrt{n} \leq \underbrace{\frac{B - \frac{1}{2}}{\sqrt{\frac{1}{12}}}}_{u_{\alpha}} \sqrt{n} \right)$$

$$G: \bar{x} \leq \frac{1}{2} + \frac{u_{\alpha}}{\sqrt{n \cdot 12}}$$

$$W = P(\bar{x} \leq B | H_1) \quad ; \quad M_{\xi} = \int_0^1 x p_1(x) dx = \frac{e-2}{e-1}$$

$$M_{\xi} = \int_0^1 x^2 p_1(x) dx = \frac{2e-5}{e-1} \quad D_{\xi} = \frac{e^2 - 3e + 1}{(e-1)^2}$$

$$P\left(\frac{\bar{x} - M_{\xi}}{\sqrt{D_{\xi}}} \sqrt{n} \leq \underbrace{\frac{B - M_{\xi}}{\sqrt{D_{\xi}}}}_{\gamma} \sqrt{n} \right) = \int_{-\infty}^{\gamma} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

$$\gamma = \frac{\frac{1}{2} + \frac{u_{\alpha}}{\sqrt{12n}} - 0.418}{\sqrt{D_{\xi}}} \sqrt{n} = \frac{\frac{u_{\alpha}}{\sqrt{12}} + 0.082 \sqrt{n}}{\sqrt{D_{\xi}}} \xrightarrow{n \rightarrow \infty} \infty$$

$W \xrightarrow{n \rightarrow \infty} 1$ т.е. асимпт. мощность

$$\alpha_1 = \alpha \quad \alpha_2 = 1 - W$$

d) $G: X_{\min} < C$

срочным критерием $\alpha_1 \leq \alpha \quad \alpha_2 \rightarrow \min$

$$\alpha_1 = P(X_{\min} < C | H_0) = 1 - (1 - F(c))^n \leq \alpha$$

где $F(x) = \int_{-\infty}^x p_0(t) dt \Rightarrow F(c) \leq 1 - \sqrt[n]{1-\alpha} \Rightarrow$

$$\Rightarrow C \leq 1 - \sqrt[n]{1-\alpha}$$

$$\alpha_2 = P(X_{\min} \geq C | H_1) = 1 - P(X_{\min} < C | H_1) =$$

$$= 1 - 1 + (1 - \Phi(c))^n = (1 - \Phi(c))^n \rightarrow \min$$

где $\Phi(c) = \int_{-\infty}^c p_1(x) dx$

для $C \in (0,1)$

$$\frac{\partial \alpha_2}{\partial C} = n (1 - \Phi(c))^{n-1} (-p_1(c)) = -\frac{n e^{1-c}}{e-1} \left(\frac{e^{1-c}-1}{e-1} \right)^{n-1} < 0$$

$$* \Phi(c) = \int_{-\infty}^c p_1(x) dx = \frac{e}{e-1} \int_0^c \frac{e^{-x}}{1} dx = \frac{e(1-e^{-c})}{e-1} *$$

$$\Rightarrow C = 1 - \sqrt[n]{1-\alpha} \Rightarrow G: X_{\min} < 1 - \sqrt[n]{1-\alpha}$$

$$\alpha_2 = (1 - \Phi(1 - \sqrt[n]{1-\alpha}))^n = \left(\frac{e^{\sqrt[n]{1-\alpha}} - 1}{e-1} \right)^n; \alpha_1 = \alpha$$

$$W = P(X_{\min} < C | H_1) = 1 - (1 - \Phi(c))^n =$$

$$= 1 - \left(\frac{e^{\sqrt[n]{1-\alpha}} - 1}{e-1} \right)^n \xrightarrow{n \rightarrow \infty} 1$$

$$* \text{ т.к. } e^{\frac{1}{n} \sqrt{1-\alpha}} = \exp \left(e^{\frac{1}{n} \ln(1-\alpha)} \right)^{\frac{1}{n}} = \exp \left(1 + \frac{1}{n} \ln(1-\alpha) + o\left(\frac{1}{n}\right) \right)$$

$$\stackrel{\text{дТ}}{=} e \left(1 + \frac{1}{n} \ln(1-\alpha) + o\left(\frac{1}{n}\right) \right) \Rightarrow W = 1 - \left(\frac{e + \frac{1}{n} \ln(1-\alpha) + o\left(\frac{1}{n}\right)}{e-1} \right)^n$$

$$\Rightarrow W = 1 - \left(1 + \frac{e \ln(1-\alpha)}{n(e-1)} \right)^n =$$

$$= 1 - e^{\frac{e \ln(1-\alpha)}{e-1}} = 1 - (1-\alpha)^{\frac{e}{e-1}} \xrightarrow{n \rightarrow \infty} 1$$

крит не явл-ся состоятельным